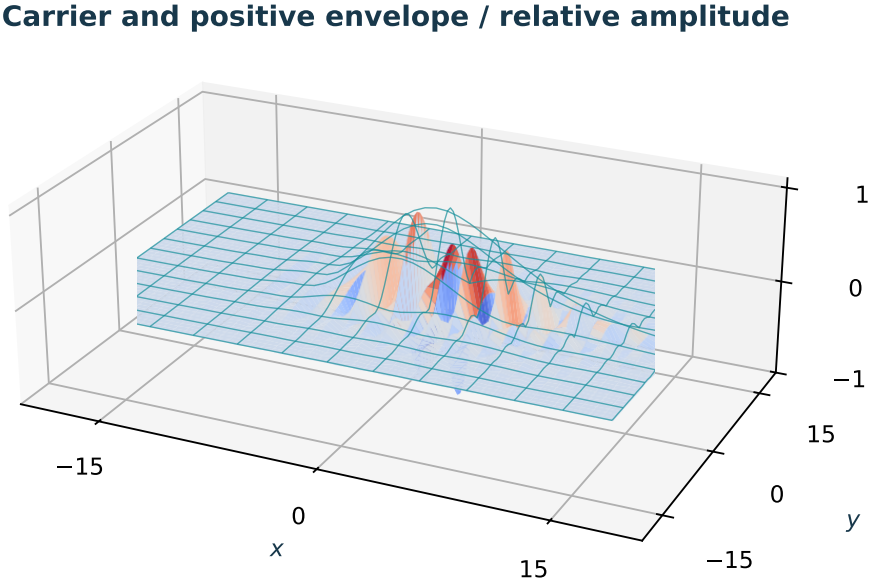
A VISUAL NUMERICAL STUDY / SEPTEMBER 2026

Waves, dispersion and decay

From individual crests to a three-dimensional decay test



A wave inside its envelope. The colored surface is an illustrative carrier reconstruction on a slice of random seed 33 at $t = 4$. The teal mesh follows its positive envelope. Height means relative amplitude; this is not a third spatial coordinate or a measured water surface.

Read this first. The latest computation gives stronger spatial support for four candidate peaks at $t = 256$: all pass a 0.1% stability screen. It does not yet establish a sharper decay theorem or a counterexample.

The reading route

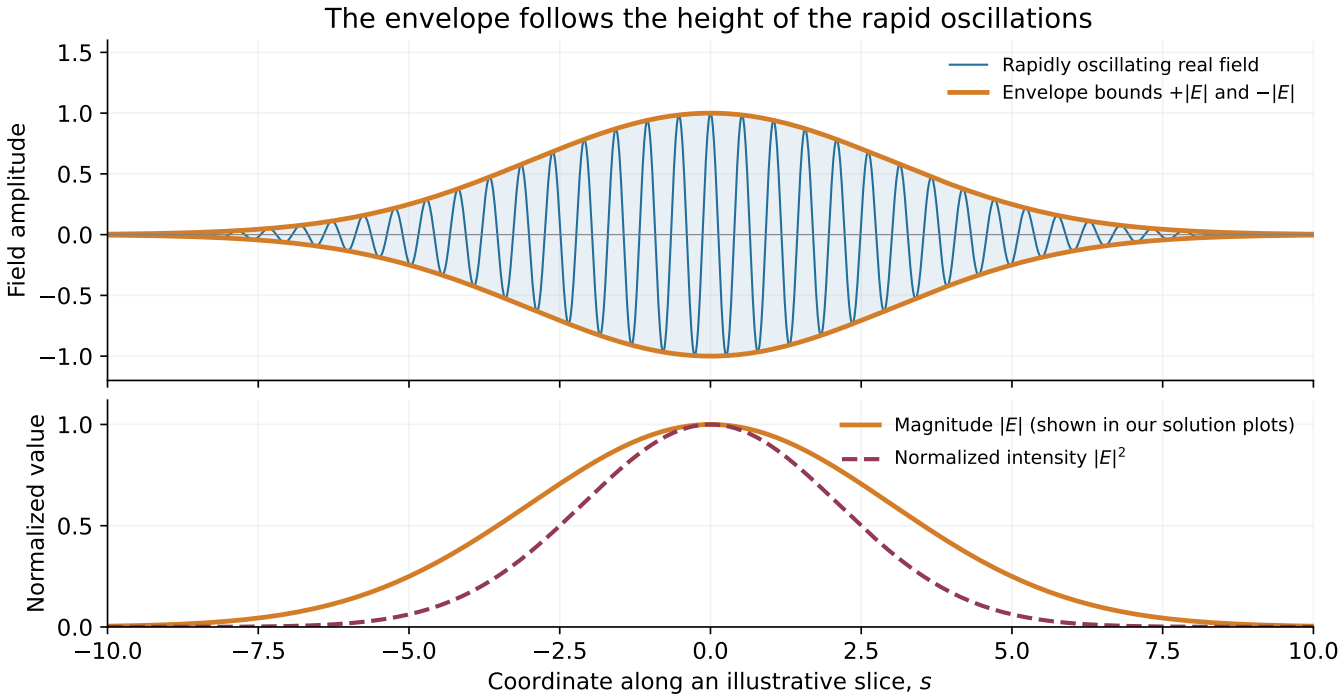
02–03 Understand the envelope and the fluid analogy.
04–06 Explore random 3D packets and the weighted diagnostic.
07–10 Read the long-time results and the broader data collection.
11–18 Inspect equations, numerical checks, and reproduction details.
19–22 Explore the new research-motivated experiments.

Scope: linear optical dispersion and canonical weighted propagators associated with the supplied Dysthe manuscript. The full nonlinear water-wave system is not simulated.

01 / READING A WAVE

The magnitude is the height of an envelope

The complex envelope carries amplitude and phase. Its magnitude tells us how large the local oscillations can be; it is not velocity. A carrier supplies the rapid crests and troughs inside that slowly varying outline.



Read this first. The oscillating curve is the wave. The upper and lower outlines are $+|E|$ and $-|E|$. A magnitude plot shows that outline, without drawing each crest.

The schematic reconstruction is

$$u_{\text{carrier}} = \text{Re}\{E\, e^{i\phi_{\text{carrier}}}\}.$$

Phase matters: two envelopes with the same magnitude can produce different crest positions and different interference. The figures on page 5 retain the simulated complex phase and add a visible carrier for illustration.

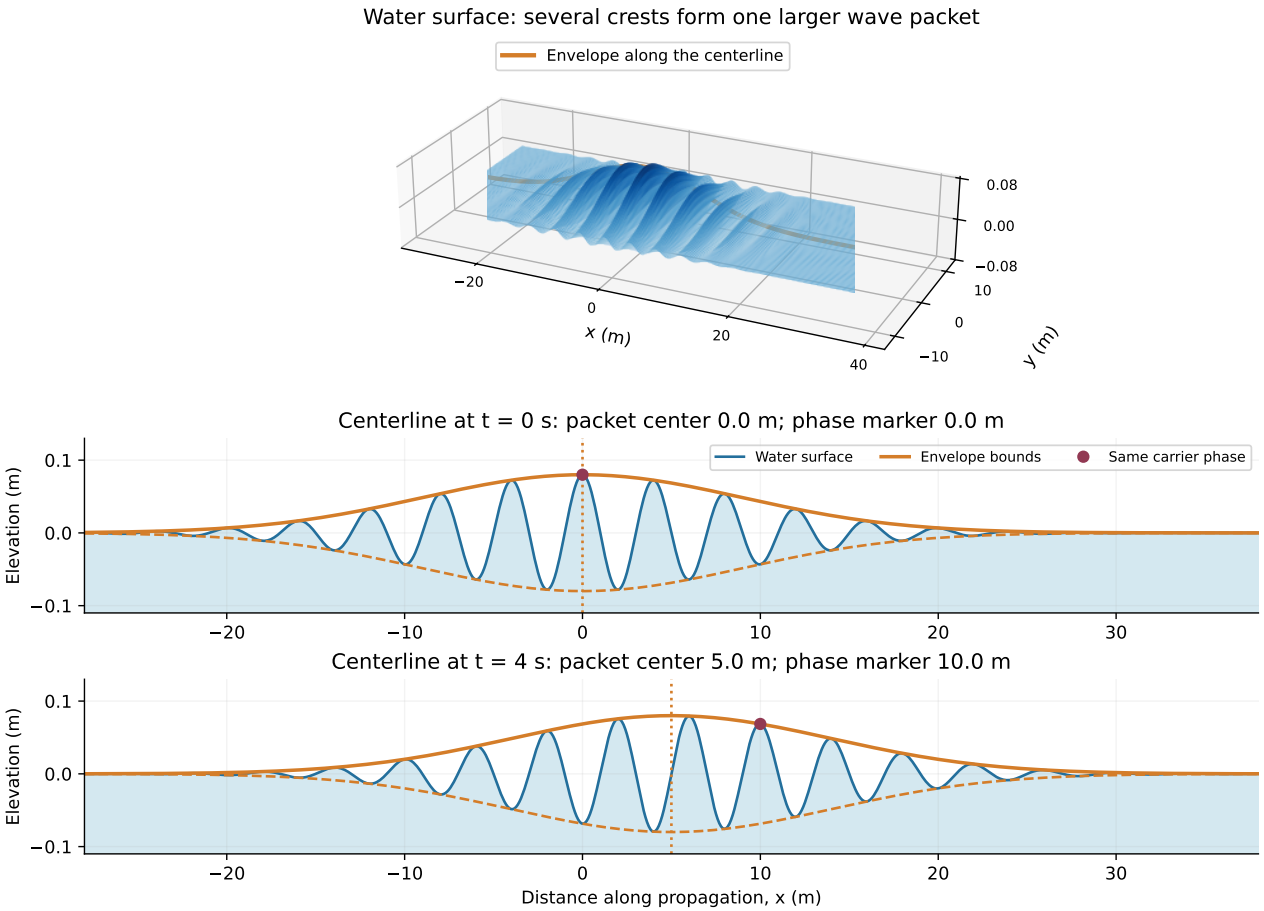
Optical intensity is proportional to $|E|^2$. A peak magnitude of 0.47 corresponds to about 0.22 of the initial peak intensity when the initial amplitude is one. Spreading can lower this peak while preserving the integral of $|E|^2$.

Units and normalization. Computations are dimensionless. The early optical Gaussian examples start at peak one. Canonical decay inputs instead have $L^1 = 1$. Their small numerical amplitudes reflect this different normalization and should not be compared directly with the optical peaks.

02 / A FLUID ANALOGY

What a packet looks like on water

A group contains several crests: they grow toward its center and fade near its edges. The envelope is an outline of their amplitude, not another material surface.



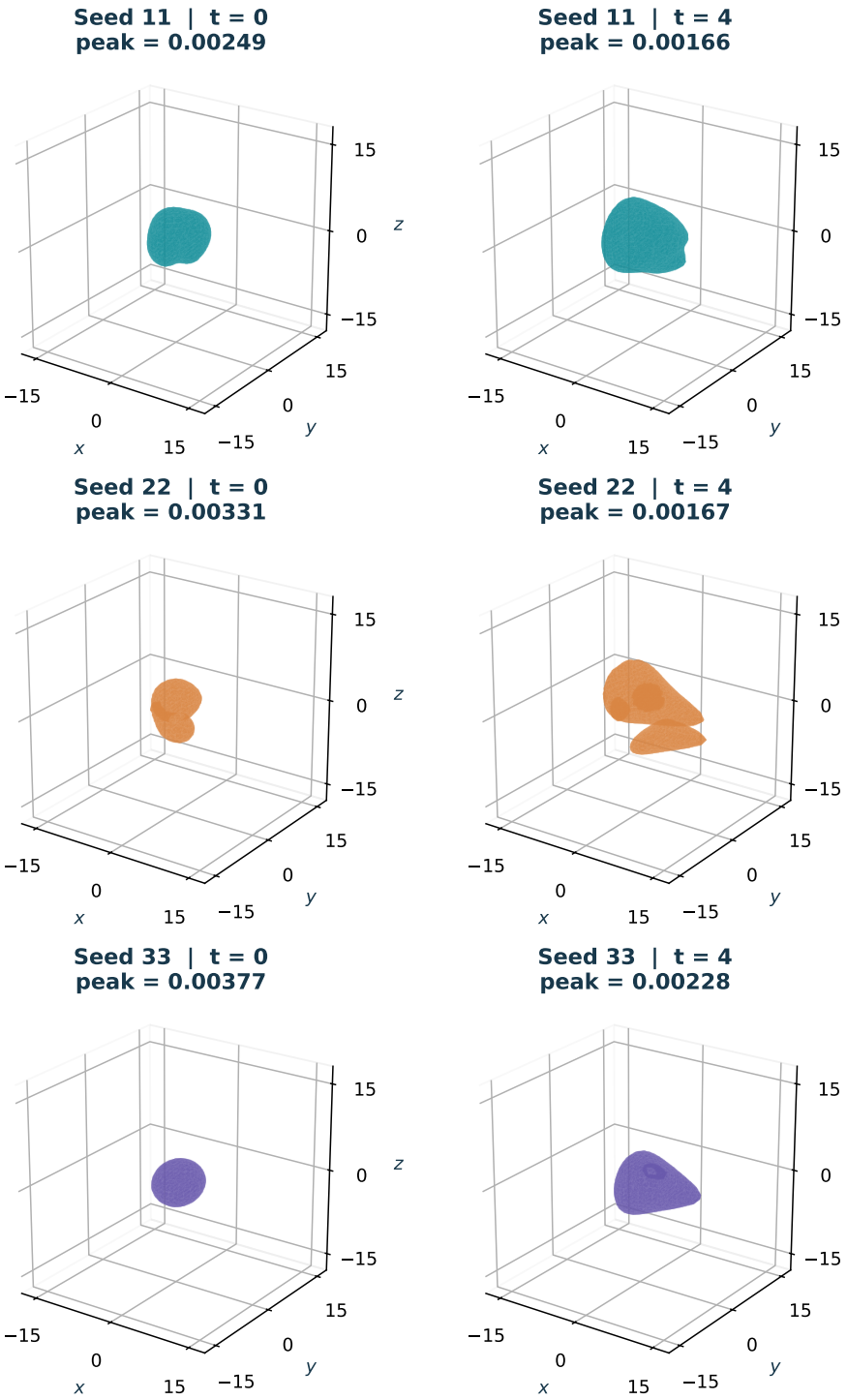
For small-amplitude deep-water gravity waves, $c_g = c_p/2$: individual crests move through the group. The dots in the lower panels track phase and group position, not water parcels. This illustration uses wavelength 4 m and amplitude 0.08 m, with vertical exaggeration. It omits spreading, nonlinear corrections and breaking.

Read this first. A surface plot has two horizontal coordinates and elevation. Our 3D field has three independent coordinates: its amplitude must be shown by color, slices, or level surfaces, as on the next page.

Fluid interpretation: R. Salmon, *Introduction to Ocean Waves*, Chapter 2, pp. 25–26. This is an explanatory analogy, not output from a fluid solver.

03 / THREE REPRODUCIBLE RANDOM CASES

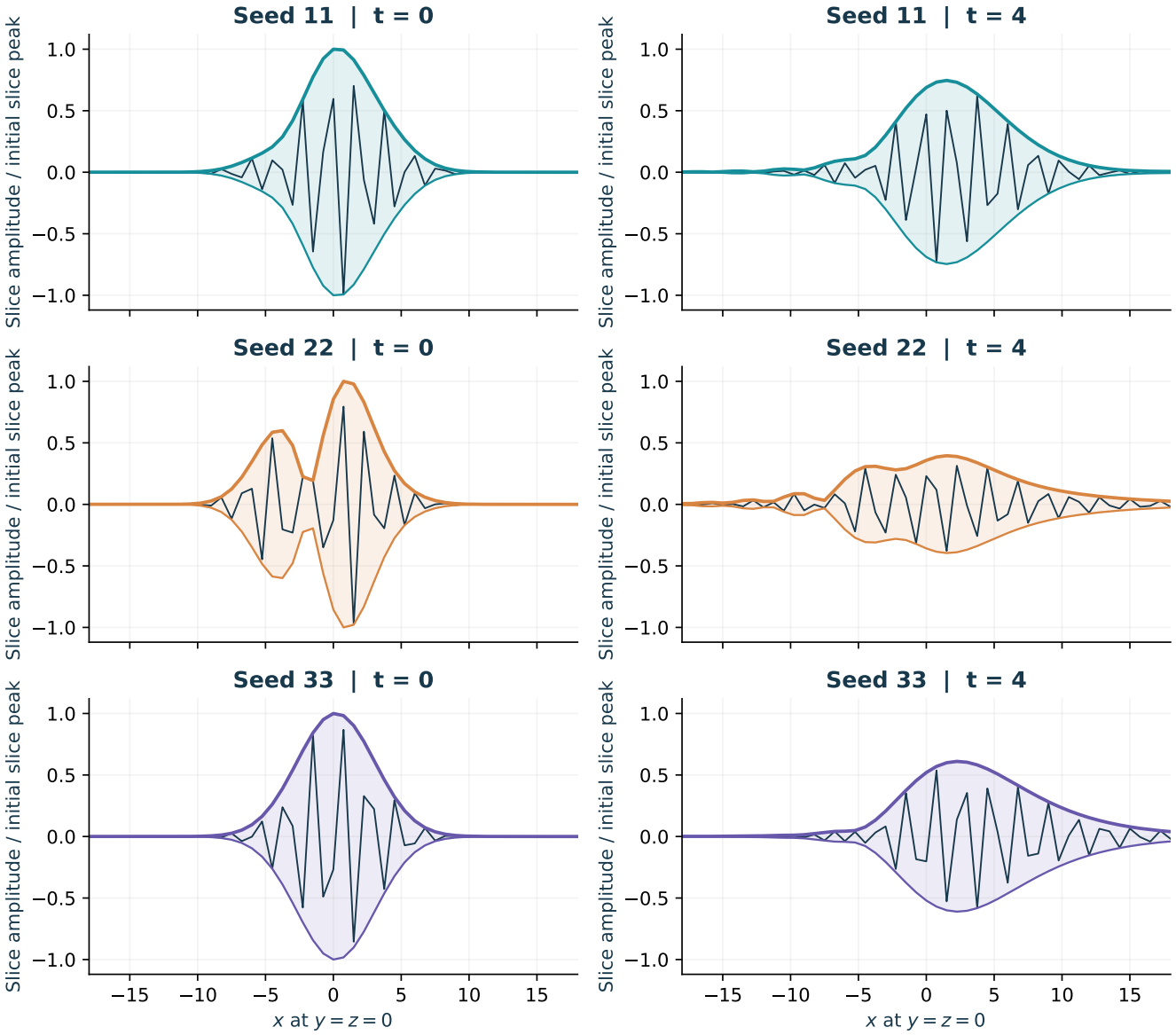
A volume can stretch, split and move



Each surface encloses $|S(t)f| \geq 0.3 \max |S(t)f|$ on the sampled volume. All axes are spatial, with the same viewing window. Thresholds follow each snapshot's own peak: compare shape here, and use the printed peaks to compare amplitude. These are three fixed seeds, not a statistical sample.

04 / LOOKING INSIDE THOSE PACKETS

The individual oscillations remain visible



The dark curves show $\mathrm{Re}\{S(t)f(x,0,0)e^{3ix}\}$; colored outlines show $\pm|S(t)f(x,0,0)|$. Each row uses its initial slice peak as a fixed divisor, so changes in amplitude remain visible between its two panels.

Read this first. An envelope does not erase individual waves. It describes their local amplitude and phase; an extra carrier reconstructs oscillations. The carrier wavenumber 3 here is chosen for readability, not inferred from physical data.

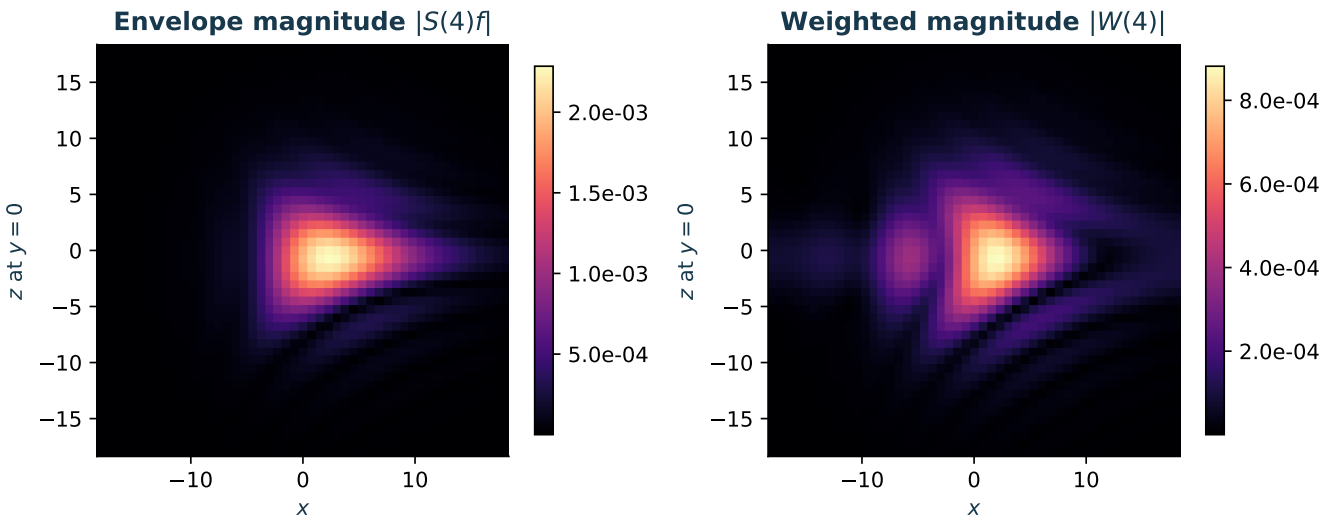
Gallery computation: homogeneous canonical symbol, 128^3 periodic grid, length 96, spacing 0.75, seeds 11/22/33. Each input is a Gaussian window times 12 random Fourier modes, normalized to discrete $L^1 = 1$. These are qualitative short-time views; the $t = 256$ spatial audit does not apply to them.

05 / FROM PICTURES TO AN INEQUALITY

Which quantity must decay?

A shrinking envelope peak is suggestive, but the paper bounds a different field: a fractional Fourier multiplier is applied before taking its maximum.

$$W(t) = |D_x + \beta|^{1/2} S(t) f, \qquad \widehat{W} = |k_x + \beta|^{1/2} e^{itw} \widehat{f}.$$



Same input, different quantity. Seed 33 at $t = 4$, on the $y = 0$ plane. Each panel has its own amplitude scale. The right panel is the field used in the inequality; it is not surface elevation or velocity.

$$\boxed{R(t; f) = \frac{t^{7/6} \|W(t)\|_\infty}{\|f\|_1} \leq C}$$

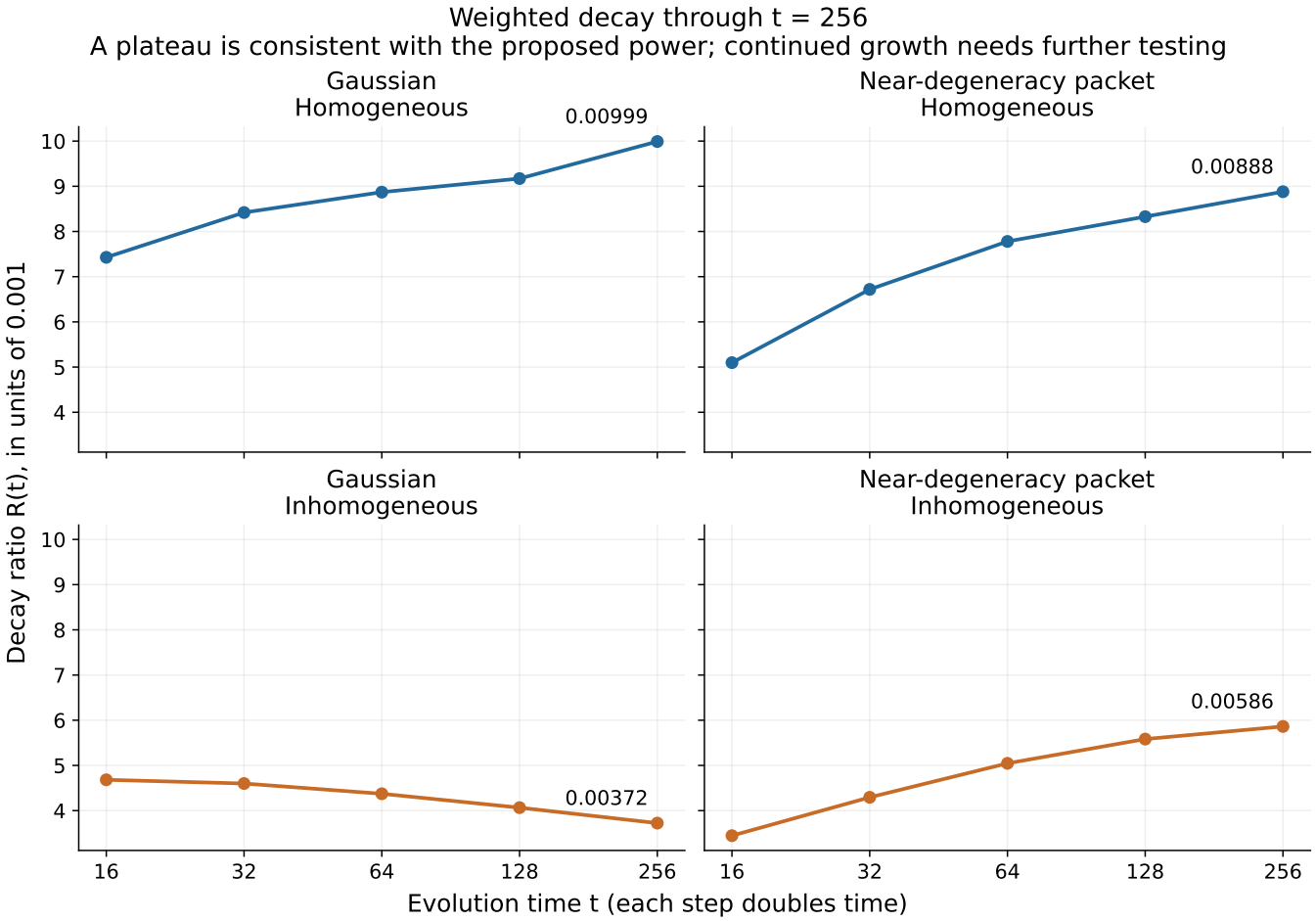
For $\alpha = 1/2$, the predicted comparison rate is $t^{-7/6}$. The constant must be independent of input and time at fixed equation parameters.

If the ratio $R(t)$ is…	The weighted peak behaves…
Flat	Like $t^{-7/6}$ over that interval.
Falling	Faster than $t^{-7/6}$ over that interval.
Rising	Slower than $t^{-7/6}$ over that interval.

Read this first. The claim is boundedness, not monotonicity. A rising finite curve can still approach a finite constant. Many inputs make the experiment more searching; they do not by themselves turn it into a uniform theorem.

06 / THE LONG-TIME RESULT

Following four cases through $t = 256$



The inhomogeneous Gaussian ratio falls at late times; the other three rise over the last doubling. Those are finite-time trends, not asymptotic conclusions.

Equation	Input	$R(128)$	$R(256)$	Local exponent
Homogeneous	Gaussian	0.00917	0.00999	1.043
Homogeneous	Near degeneracy	0.00833	0.00888	1.074
Inhomogeneous	Gaussian	0.00407	0.00372	1.293
Inhomogeneous	Near degeneracy	0.00558	0.00586	1.096

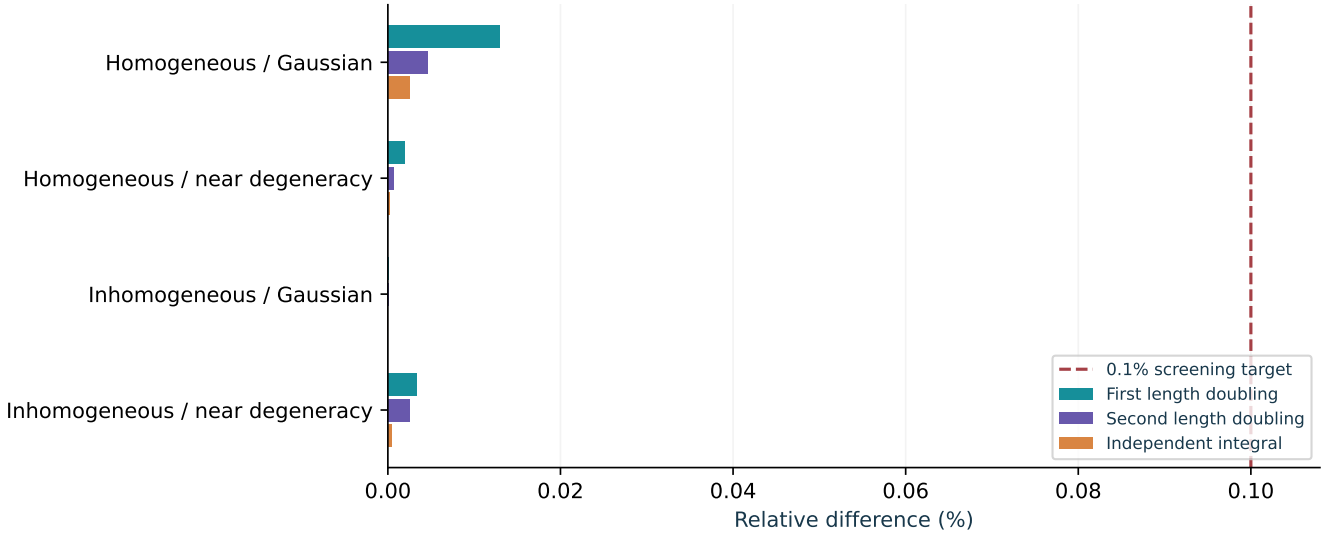
The exponent is $-\log[M(256)/M(128)]/\log 2$, where $M = \|W\|_\infty$. Compare it with $7/6 \simeq 1.167$; it describes one time doubling only. These curves and entries retain the long-time batch values. The subsequent audit refines the four endpoint ratios in the technical table at the end of this report.

Read this first. We now have a longer observation window. The strongest result is the improved reliability of its four endpoint peaks, shown next. No counterexample or stronger uniform exponent has been established.

07 / CAN WE TRUST THE SPATIAL COMPUTATION?

Enlarge the search, then compare again

At fixed $t = 256$, we increased the longitudinal period from 32,768 to 65,536 and then 131,072. Separately, the radial search grew from 2,048 to 4,096. The broad search retained $\Delta x = 0.25$ and $\Delta r = 2$, followed by local peak optimization.



Read this first. All four cases pass the 0.1% numerical screen. The largest final change in refined peak magnitude is 0.00459%; the largest location drift is 0.00079 spatial units.

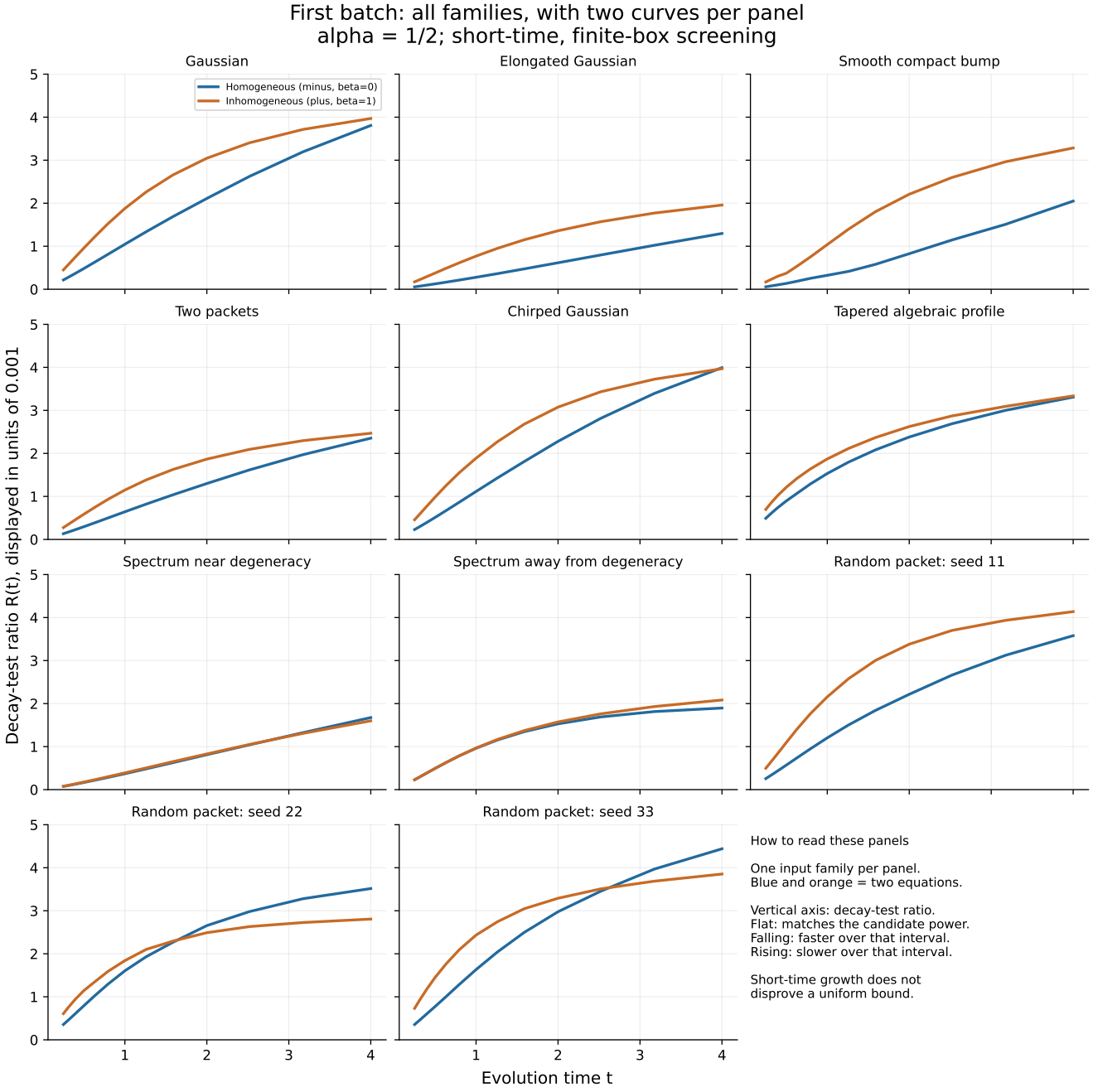
An independent non-periodic integral agrees within 0.0026% at the candidate peaks. Extending the radial search found no larger sampled maximum. These checks support stability under the tested changes, rather than supplying a rigorous error bound.

Why length matters. The inhomogeneous near-degeneracy peak is near $x = -841$ at $t = 256$. It lies far outside the length-288 box used for the earlier experiments. A long-time curve on that old box could confuse periodic recirculation with whole-space behavior.

What remains unresolved. A finite search and local refinement can miss a narrow or distant maximum. This audit covers four Gaussian-based cases at one time and one derivative weight. It does not transfer automatically to random inputs.

08 / BEYOND A SINGLE GAUSSIAN

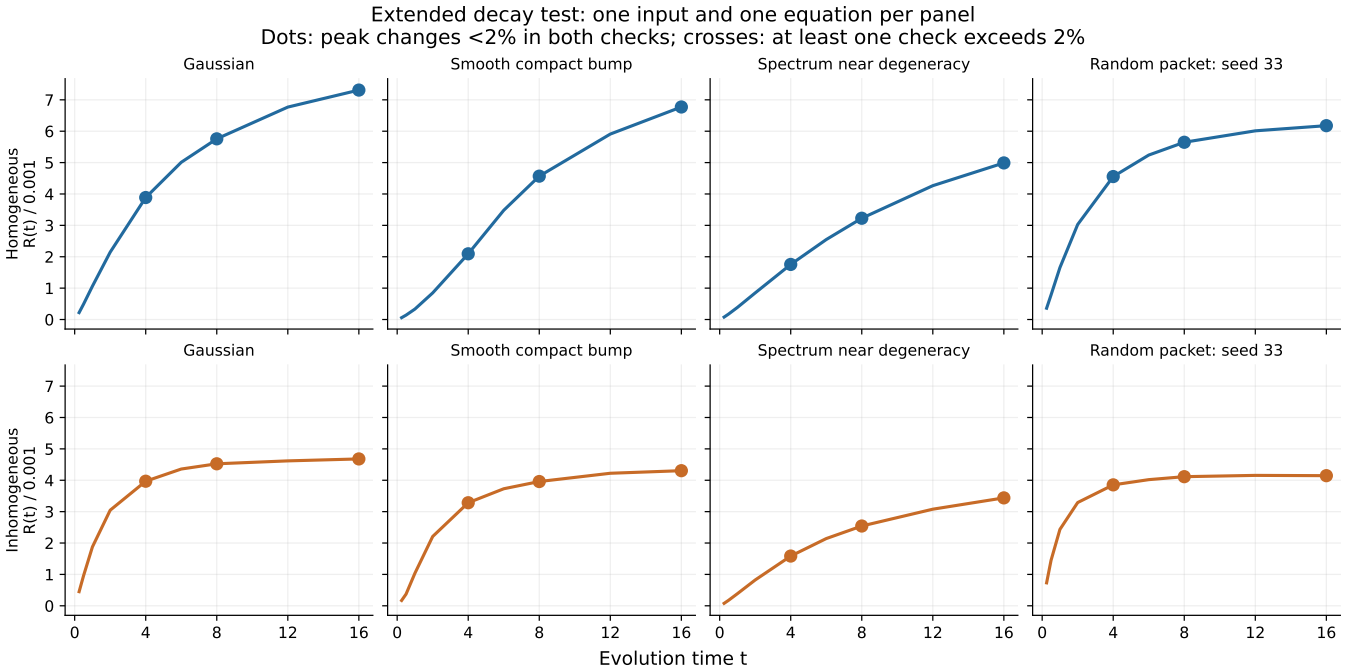
Eleven families make the test more varied



Original short-time survey: two equation regimes per panel, $\alpha = 1/2$, length-96 box. These curves remain exploratory finite-box measurements. Recipes include compact, anisotropic, chirped, tapered algebraic and random inputs. The three random seeds shown earlier belong to this collection. No measured fluid dataset has been imported.

09 / A MORE CAREFUL INTERMEDIATE WINDOW

Four families checked through $t = 16$



Curves use a length-288 cube at spacing 1. Dots identify the checked times $t = 4, 8, 16$ passing the exploratory 2% peak-change screen in both domain and grid comparisons; intermediate times were not independently checked.

Read this first. Across 24 checked case–time combinations, the largest domain peak change is 1.62% and the largest grid peak change is 0.94%. This supports the intermediate-time comparisons, with weaker tolerances than the four-case endpoint audit.

Next scientific step. Extend the reliable time window for more recipes, keeping input normalization fixed and testing spatial stability at each new stage. A candidate contradiction needs sustained growth that survives those checks; strengthening a uniform theorem needs an argument covering its full input class.

Technical record follows. The remaining pages retain the equations, optical baseline results, detailed refinement tables and reproducible algorithms. The optical plots use an unweighted field and a different normalization from the canonical ratio experiments; they explain dispersion but are not direct bound tests.

TECHNICAL RECORD / OPTICAL BASELINE

1. Equation and numerical model

We solve

$$iE_t + (1 - i\varepsilon_1\partial_z)\Delta_\perp E - E_{zz} - i\varepsilon_2 E_{zzz} = 0, \qquad \Delta_\perp = \partial_x^2 + \partial_y^2. \tag{1}$$

With $\widehat{E}(\mathbf{k}) = \int E(\mathbf{x})e^{-i\mathbf{k}\cdot\mathbf{x}}\,d\mathbf{x}$,

$$\widehat{E}(\mathbf{k}, t) = e^{-it\omega(\mathbf{k})}\widehat{E}_0(\mathbf{k}), \qquad \omega = (1 + \varepsilon_1 k_z)(k_x^2 + k_y^2) - k_z^2 + \varepsilon_2 k_z^3. \tag{2}$$

This matches the paper's symbol (6.6). Its equation (1.32) prints the opposite sign on the mixed third derivative. We follow (1.27) and (6.6) consistently; this choice must be retained when comparing with another implementation.

An FFT computes the initial coefficients once. For each requested t, we multiply by the phase in (2) and perform an inverse FFT. Evolution of the retained modes is exact up to floating-point arithmetic. There is no time-marching truncation error; spatial sampling, finite-box effects, and sampling the peak remain relevant. At a fixed total grid size N, M snapshots cost $O(MN \log N)$.

Setting	Value
Computational box	$[-48, 48)^3$, periodic
Production grid	96^3 points; $\Delta x = \Delta y = \Delta z = 1$
Arithmetic	Torch CPU, complex128, four threads
Evolution times	25 snapshots, $t = 0, 0.25, \ldots, 6$
Parameters	$\varepsilon_1 = 1$; $\varepsilon_2 \in \{1, -1/3, 0\}$
Initial Gaussian	$E_0 = e^{-(x^2+y^2+z^2)/18}e^{i\mathbf{q}\cdot\mathbf{x}}$
Frequency shifts	$\mathbf{q} = 0$, $(\pi/6, 0, 0)$, $(0, 0, \pi/6)$

The shift is eight Fourier bins on the production box. All three initial magnitudes are identical, with peak one. Positive ε_2 and $-\varepsilon_1/3$ connect to the parameter cases in Theorem 5.1; the zero case is included as a linear comparison, without extending that nonlinear theorem to it. All quantities are dimensionless.

2. Peak evolution and parameter dependence

The peak curves below are spatial grid maxima, not amplitudes at a fixed point. All three parameter families spread over the observation window. The response to a z-shift depends strongly on ε_2, even though the initial magnitudes and norms are unchanged.

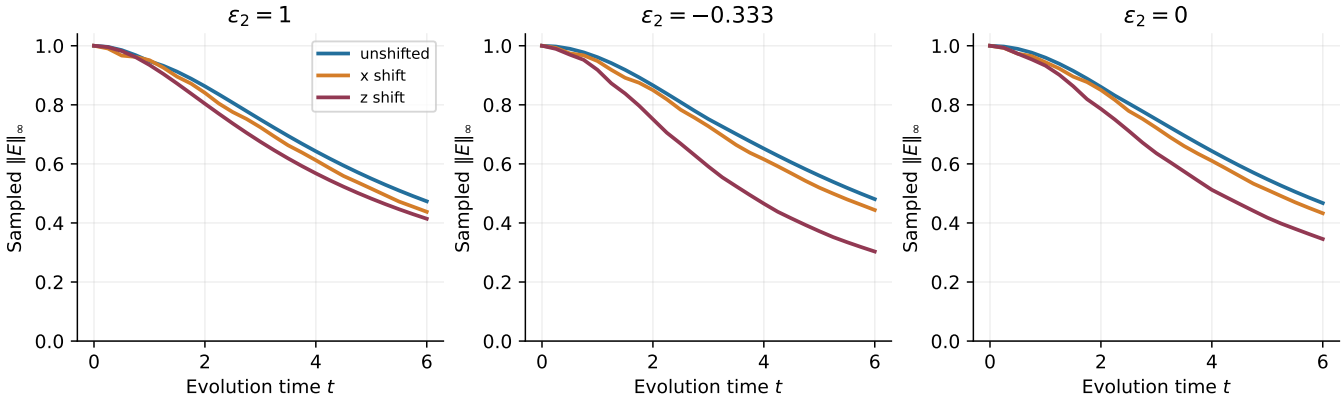


Figure 3. Sampled L^∞ histories. Lines join directly evaluated snapshots; they do not represent a numerical time integrator.

ε_2	Initial spectrum	Peak at $t = 6$	s_x	s_y	s_z
1	unshifted	0.473	3.598	3.598	3.866
1	x shift	0.438	3.891	3.598	4.140
1	z shift	0.414	4.849	4.849	3.090
$-1/3$	unshifted	0.481	3.598	3.598	3.629
$-1/3$	x shift	0.444	3.891	3.598	3.919
$-1/3$	z shift	0.303	4.849	4.849	4.872
0	unshifted	0.467	3.598	3.598	3.598
0	x shift	0.433	3.891	3.598	3.891
0	z shift	0.346	4.849	4.849	3.598

Here s_j is the standard deviation of coordinate j under the normalized intensity $|E|^2 / \int |E|^2$. Initially $s_x = s_y = s_z = 3/\sqrt{2} \simeq 2.121$. Widths describe the packet as a whole and remain meaningful when its maximum moves.

For $\varepsilon_2 = 1$, the final peaks are approximately 0.473 (unshifted), 0.438 (x-shift), and 0.414 (z-shift). For $\varepsilon_2 = -1/3$, the z-shifted peak is approximately 0.303. The z-shift therefore need not have the same effect across dispersion parameters.

What these curves establish. They demonstrate finite-time spreading and directional sensitivity for these initial data. They do not establish an asymptotic decay exponent, monotone decay for arbitrary data, or a universal decay constant. The reported L^2 norm is constant to relative error 1.11×10^{-16}; this is dispersion without imposed damping.

3. Spatial solutions

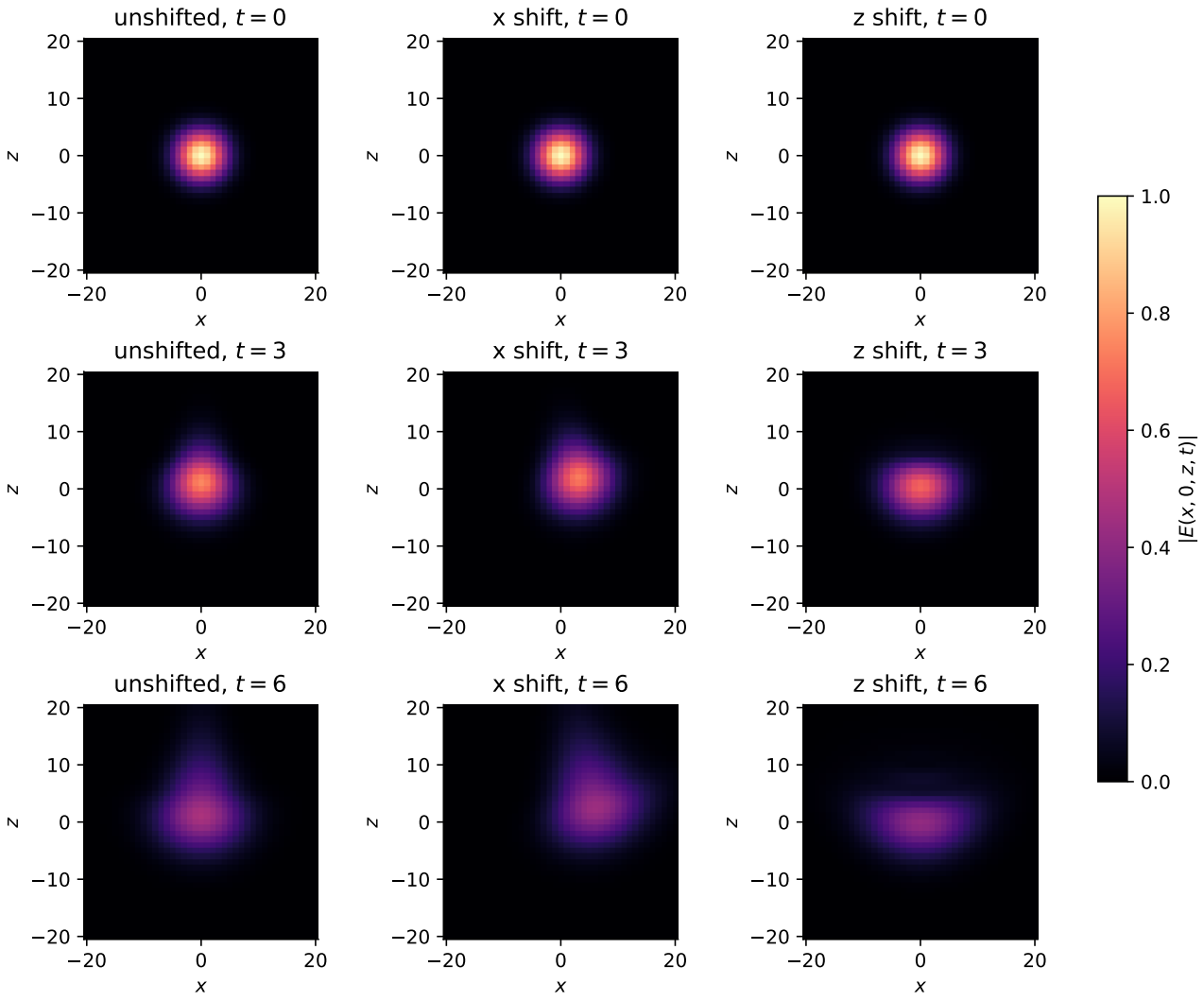


Figure 4. Magnitude on the $y = 0$ plane for $\varepsilon_2 = 1$. Columns compare unshifted, x-shifted, and z-shifted initial spectra. Rows show $t = 0, 3, 6$. Every panel uses the same color range, making the loss of peak amplitude visible. The displayed window is $[-20, 20]^2$ within the larger computational box; the color plots are slices of genuine three-coordinate solutions.

Modulation changes propagation as well as deformation. A slice alone can miss an off-plane maximum, which is why Figure 3 and the table use the full volume. Full complex fields at $t = 6$ are supplied in the accompanying snapshot archive.

4. Directional spreading and generic initial data

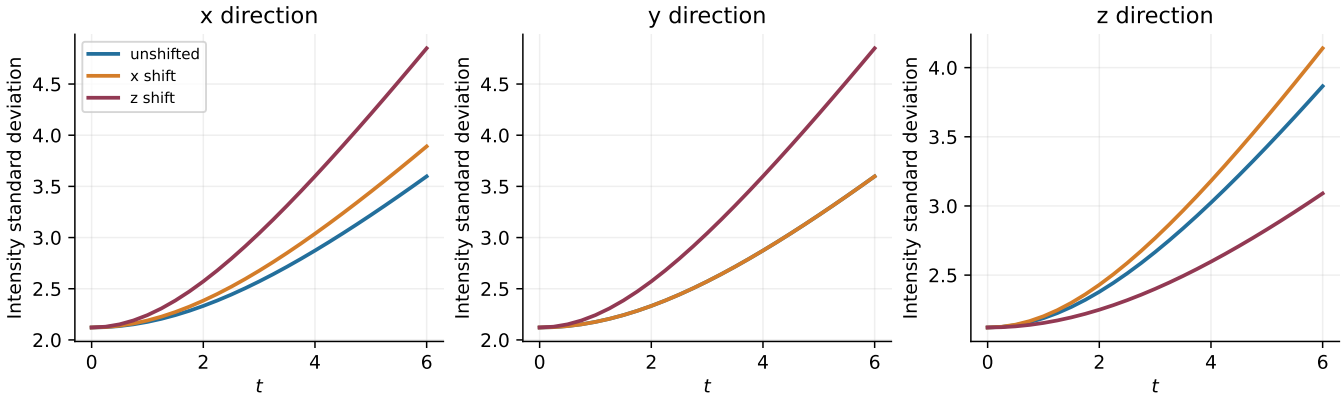


Figure 5. Intensity widths for $\varepsilon_2 = 1$. An x-shift breaks the equality of the transverse widths; a z-shift changes both transverse widths together. Exchanging x and y in the shifted experiment agrees to maximum absolute field error 1.76×10^{-16}.

This behavior is consistent with the group velocity of (2):

$$\nabla_{\mathbf{k}}\omega = \big(2(1+\varepsilon_1 k_z)k_x,\ 2(1+\varepsilon_1 k_z)k_y,\ \varepsilon_1(k_x^2+k_y^2) - 2k_z + 3\varepsilon_2 k_z^2\big).$$

Changing k_z changes the transverse dispersion factor itself. Changing k_x also affects longitudinal transport through the k_x^2 term. The axes are coupled.

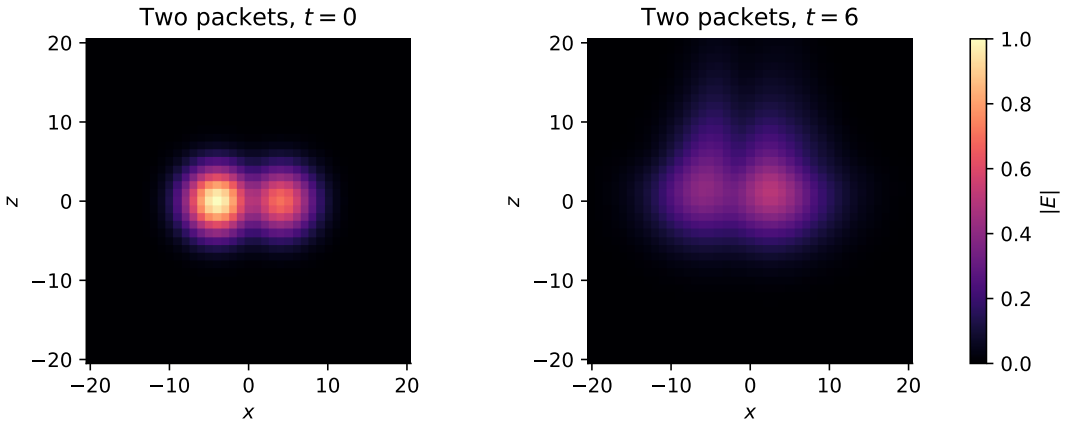


Figure 6. A non-Gaussian initial profile, propagated with the same solver:

$$E_0 = e^{-((x+4)^2+y^2+z^2)/18} + 0.7i\, e^{-((x-4)^2+y^2+z^2)/18}, \qquad \varepsilon_2 = 1.$$

This example demonstrates that the solver accepts a sampled complex field without requiring a Gaussian formula. Its separate convergence study is not included here.

5. Verification and interpretation limits

Algebraic checks. A periodic plane wave with wavevector $(1, 2, -1)$ at $\varepsilon_1 = \varepsilon_2 = 1$ has $\omega = -2$, hence exact multiplier e^{2it}. The solver agrees at $t = 0.73$ to 2.18×10^{-15} in maximum error. Forward/backward propagation of the z-shifted packet gives relative L^2 error 6.69×10^{-16}. These checks complement conservation, which alone cannot detect a wrong phase sign.

Grid refinement. The table compares the z-shifted field at $t = 6$ against a 192^3 computation on the same length-96 box. The reference is restricted to the coarse nodes before measuring relative discrete L^2 error.

Grid	$\varepsilon_2 = 1$	$\varepsilon_2 = -1/3$	$\varepsilon_2 = 0$
32^3	1.89×10^{-1}	2.05×10^{-1}	1.68×10^{-1}
64^3	7.75×10^{-6}	8.24×10^{-6}	7.38×10^{-6}
96^3	2.13×10^{-14}	2.15×10^{-14}	1.90×10^{-14}

This demonstrates convergence of sampled fields on a fixed periodic box. It does not imply that a grid maximum equals the continuum supremum. Across all nine final Gaussian fields, doubling the resolution changes the sampled peak by at most 0.556 percent. Figure 3 retains the production-grid maxima.

Domain enlargement. At spacing 0.5, length-48 and length-72 boxes are compared to length 96 on the common cube $[-24, 24)^3$, using the z-shifted field. A separate check compares each production field to a length-144 box at spacing 1, restricted to the entire original box. The last column is the worst of all three shifts.

ε_2	$L = 48$ vs. 96	$L = 72$ vs. 96	$L = 96$ vs. 144
1	5.87×10^{-3}	2.00×10^{-5}	2.07×10^{-4}
$-1/3$	3.70×10^{-2}	1.71×10^{-5}	1.55×10^{-5}
0	3.95×10^{-3}	1.49×10^{-7}	5.58×10^{-7}

Values are relative discrete L^2 differences on the stated comparison regions. The agreement improves on enlarging the box, but it is evidence on $0 \le t \le 6$, not permission to extrapolate indefinitely on a periodic domain.

Independent whole-space check. For the Gaussian, the transverse Fourier integrals can be evaluated analytically, leaving a one-dimensional adaptive integral in k_z. At $t = 6$, we compare the origin, $(3, -2, 5)$, and the production peak node for each of the nine cases. The largest absolute complex-field discrepancy is 3.91×10^{-9}. Quadrature uses $k_z \in [-5, 5]$ with absolute and relative tolerance 10^{-12}; Gaussian tails outside this interval are negligible at this precision. This checks 27 points, not the entire continuum field.

Relation to the paper. Propositions 6.2–6.3 bound a fractional-derivative-weighted solution in transformed coordinates for $0 < \alpha < 1$. They do not directly assert the plotted unweighted peak follows $t^{-(3+\alpha)/3}$. No exponent is fitted here, and no theorem constant is inferred. The weighted experiments earlier in this combined report address that separate diagnostic.

Reproduction. From the project folder, run the following commands:

```
python3 numerical_solutions.py
python3 report/make_tables.py
```

Compile `unified_report.tex` from `report/`. Dependencies are NumPy, SciPy, Matplotlib, and Torch. The supplied CSV contains all 225 Gaussian norm records; JSON contains checks and configuration; the NPZ contains nine final Gaussian fields and the two-packet field. The recorded computation and figure-generation time was 12.4 seconds on this host; this is one CPU run, not a hardware benchmark or a GPU measurement.

Source. R. Mosincat, D. Pilod, and J.-C. Saut, *On Third Order Nonlinear Schrödinger Equations*, supplied manuscript `Dysthe3D-2.pdf`, pp. 9–10 and 18–23. No publication status is assumed.

TECHNICAL RECORD / CANONICAL PROPAGATOR

Weighted equation and intermediate checks

We use $S(t)f = \mathcal{F}^{-1}(e^{itw}\widehat{f})$ with

$$w = k_x^3 + 3sk_x(k_y^2 + k_z^2) + 3\beta(k_y^2 + k_z^2),$$

where $(s, \beta) = (-1, 0)$ or $(1, 1)$. This follows the proofs of Propositions 6.2–6.3; the signs printed in (6.1) are inconsistent. The weighted multiplier is $|k_x + \beta|^{1/2}$. These are canonical coordinates, not a numerical conversion of the optical baseline above.

We extend four families in both regimes to $t = 16$, four times the first batch's maximum time. Gaussian is the reference, the compact bump has a broader spectrum, the near-degeneracy packet targets the multiplier's cusp, and seed 33 represents the random collection. This is a targeted follow-up, not an exhaustive data search.

Domain checks compare lengths 96, 192, and 288 at spacing 1. A separate length-192 grid with spacing 2/3 checks resolution. Every comparison uses the same continuum recipe, normalized using its own grid quadrature. The table gives results at $t = 16$; machine-readable files also include $t = 4, 8$ and complex-field errors on a common cube.

Equation	Input	$R(16)$	Domain peak change	Grid peak change
Homogeneous	Gaussian	0.00731	1.11%	0.18%
Homogeneous	Compact bump	0.00677	0.43%	0.03%
Homogeneous	Near degeneracy	0.00499	1.62%	0.24%
Homogeneous	Random (33)	0.00618	1.61%	0.29%
Inhomogeneous	Gaussian	0.00468	0.01%	0.05%
Inhomogeneous	Compact bump	0.00431	0.00%	0.54%
Inhomogeneous	Near degeneracy	0.00344	1.36%	0.01%
Inhomogeneous	Random (33)	0.00415	0.00%	0.04%

Domain peak change compares lengths 192 and 288; grid peak change compares spacings 1 and 2/3 at length 192. Peak comparisons use each grid's full sampled maximum, not a fixed spatial point. The plotted length-288 field is not itself also refined to spacing 2/3, so the checks do not establish joint convergence.

Observed behavior. The inhomogeneous Gaussian and random packet ratios begin to flatten: from $t = 8$ to 16 they grow by 3.42% and 0.78%, respectively. The near-degeneracy ratios continue to rise more visibly, as do several homogeneous cases. This suggests different transient time scales; it does not establish their eventual behavior.

Numerical qualification. Across all 24 checked case–time combinations, the largest peak changes are 1.62% under domain enlargement and 0.94% under grid refinement. All pass the exploratory 2% screen.The fractional multiplier can produce long spatial tails, and errors can vary nonmonotonically with box size near its cusp. A field's relative L^2 difference and its maximum-amplitude difference are distinct: this follow-up prioritizes the maximum because it appears in the inequality, while retaining both diagnostics.

Inference. These experiments do not establish a faster universal decay rate, a theorem constant, or a counterexample. A convincing candidate violation would require a controlled sequence of times or normalized inputs with increasing ratios that survives numerical refinement. A short transient slope is insufficient. Cases that fail the peak checks should be refined before being interpreted as evidence about the whole-space bound.

Data and reproducibility. Run `python3 decay_dataset.py`, then `python3 refine_decay.py`. Results are in `results/decay_dataset/`: `refined_decay.csv`, `refinement_checks.json`, `REFINEMENT.md`, and the plotted figures. The first batch's inputs remain available as complex-valued NPZ arrays. These are artificial inputs. CDIP buoy observations have been identified as a possible source, but are not imported or presented as three-dimensional measurements.

Source. R. Mosincat, D. Pilod, and J.-C. Saut, *On Third Order Nonlinear Schrödinger Equations*, supplied manuscript `Dysthe3D-2.pdf`, Section 6. This supplement follows the stated estimates as research targets, without claiming to resolve the manuscript's proof details.

Visual gallery reproduction. Run `python3 report/make_visual_gallery.py`. The script records seeds, grid, normalization, peak values and L^2 norms in `report/visuals/gallery_manifest.json`. Rebuild this combined source with `python3 report/build_unified.py`, then compile `report/unified_report.tex`.

How the long-time calculation is checked

Let $r = \sqrt{y^2 + z^2}$ and use the exactly L^1-normalized initial profile

$$f(x, r) = \frac{\exp[-x^2/(2\sigma_x^2) - r^2/(2\sigma_y^2)]e^{iqx}}{(2\pi)^{3/2}\sigma_x\sigma_y^2}.$$

The Gaussian has $(\sigma_x, \sigma_y, q) = (3, 3, 0)$; the near-degeneracy packet has $(6, 3, -\beta)$. Analytically integrating both transverse Fourier variables reduces the weighted field to

$$W(x, r, t) = \frac{1}{4\pi^2}\int_{\mathbb{R}} \frac{|k+\beta|^{1/2}}{d(k,t)} \exp\left[-\frac{\sigma_x^2(k-q)^2}{2} + i(tk^3 + kx) - \frac{r^2}{2d(k,t)}\right] dk, \qquad d(k,t) = \sigma_y^2 - 6it(sk + \beta).$$

This is the same three-coordinate linear solution for these inputs. Only the evaluation method changes. The transverse directions are treated on the whole space; a longitudinal FFT quadrature evaluates many x locations efficiently. Radial symmetry reduces the spatial peak search to (x, r).

Check	x period	Δx	Max. r	Δr
Base	4096	0.5	512	4
Expanded	8192	0.5	1024	4
Refined	8192	0.25	1024	2
Extra expansion at $t = 256$	32768	0.25	2048	2

The full (x, r) search is followed by local optimization of the peak. At 256, the extra expansion changes the sampled peak by at most 0.141%. An independent adaptive quadrature of the integral above agrees with the complex field at each final candidate peak to within 0.020% relative error. The largest sampled search-boundary amplitude is 0.586% of its case's peak. These checks are numerical evidence, not a certification that no larger unsampled peak exists.

Why the previous box was inadequate. At $t = 256$, the inhomogeneous near-degeneracy maximum lies around $x = -841$. It would be outside even the length-288 box used for the earlier $t \le 16$ runs. Extending time without changing the spatial computation would therefore be unreliable.

Remaining limits. The Gaussian-specific reduction cannot be applied unchanged to arbitrary measured fields or every artificial recipe. The adaptive quadrature checks selected peak locations; the global search remains finite and sampled. The two-dimensional radial search, local optimizer status, boundary ratios, and domain/resolution comparisons are all recorded to make these limits reviewable.

Reproduce. Run `python3 long_time_decay.py`, followed by `python3 report/make_long_time_tables.py`. Results and the graph are under `results/long_time/`; compile `report/unified_report.tex` for this combined numerical study. Source: the same manuscript, canonical symbols (6.2) and weighted estimates in Propositions 6.2–6.3.

Stronger spatial reliability at $t = 256$

Purpose. Hold time, input, and $\alpha = 1/2$ fixed, and vary the spatial computation. This audit separates longitudinal enlargement from transverse enlargement. It tests whether the four reported candidate maxima stabilize to an exploratory relative tolerance of 0.1%.

The longitudinal period increases from 32,768 to 65,536 while the radial limit stays at 2,048. We then search the additional annulus out to radius 4,096. Finally, we double the longitudinal period again to 131,072, retaining the larger radial limit. Every broad scan keeps $\Delta x = 0.25$ and $\Delta r = 2$. The baseline is the saved final scan from the preceding experiment.

Equation	Input	First L change	Radius change	Second L change	Integral check
Homogeneous	Gaussian	0.0130%	0.0000%	0.0046%	0.0025%
Homogeneous	Near degeneracy	0.0020%	0.0000%	0.0007%	0.0003%
Inhomogeneous	Gaussian	0.0001%	0.0000%	0.0000%	0.0000%
Inhomogeneous	Near degeneracy	0.0033%	0.0000%	0.0026%	0.0004%

The three domain columns compare sampled peak magnitudes before and after the stated expansion. The integral check compares the final Fourier-quadrature field to an independently evaluated non-periodic integral at the refined candidate peak. Percentages use the newer or independent result as the denominator. Values are rounded.

Result. 4 of four cases pass the 0.1% peak screen. This includes the two longitudinal expansions, the transverse expansion, the final locally optimized peak drift, and the independent integral check. The largest change in locally optimized peak magnitude between the final two periods is 0.00459%.

Equation	Input	Refined $R(256)$	x	r	Location drift
Homogeneous	Gaussian	0.0099887	-20.7469	0.0000	0.00000
Homogeneous	Near degeneracy	0.0088803	7.7961	6.5381	0.00079
Inhomogeneous	Gaussian	0.0037238	-8.9154	0.0000	0.00002
Inhomogeneous	Near degeneracy	0.0058621	-841.2464	0.0000	0.00021

Locations are continuous-coordinate candidate maxima from the independent integral optimization. The drift is the Euclidean distance in (x, r) between local peak optimizations at the final two longitudinal periods. Its maximum is 0.00079. Axial symmetry means a nonzero r represents a ring of equivalent points, not a unique three-dimensional location.

Additional checks. Local optimization starts from the broad search's largest sample and refines both x and r. Increasing the independent integral's Gauss–Legendre order from 256 to 512 per panel changes the complex field by at most 3.67×10^{-12}% relative to the tighter result. Its integration interval is split at the fractional multiplier cusp and into 64 Gaussian-frequency panels. On each side, $k = -\beta \pm u^2$ removes the square-root cusp from the integration variable. Fixed-order quadrature with an order-doubling check replaces adaptive quadrature, which issued convergence warnings during local searches. The largest sampled search-boundary amplitude is 0.00516% of the corresponding peak. Optimizer status and all intermediate measurements are saved in the per-case JSON records.

What we can now say. At this time and for these four inputs, the peak values have stronger numerical support from separate domain enlargements, continuous local refinement, stable locations, and independent integration. This improves confidence in the reported decay ratios. It does not prove the maximum is global over all of $\mathbb{R}^3$, establish an asymptotic decay rate, or extend reliability to other inputs, values of α, or later times.

Remaining spatial limitation. The broad radial mesh is still spaced by 2; refinement is local to the identified maximum, not a global halving of every cell. A missed narrow peak or a remote maximum is not excluded by these checks. The report therefore claims numerical spatial stability of the identified peaks, not a certified supremum.

Reproduce. Run `python3 spatial_reliability.py`, then `python3 report/make_spatial_tables.py`. Results, configurations, and checkpoints are under `results/spatial_reliability/`.

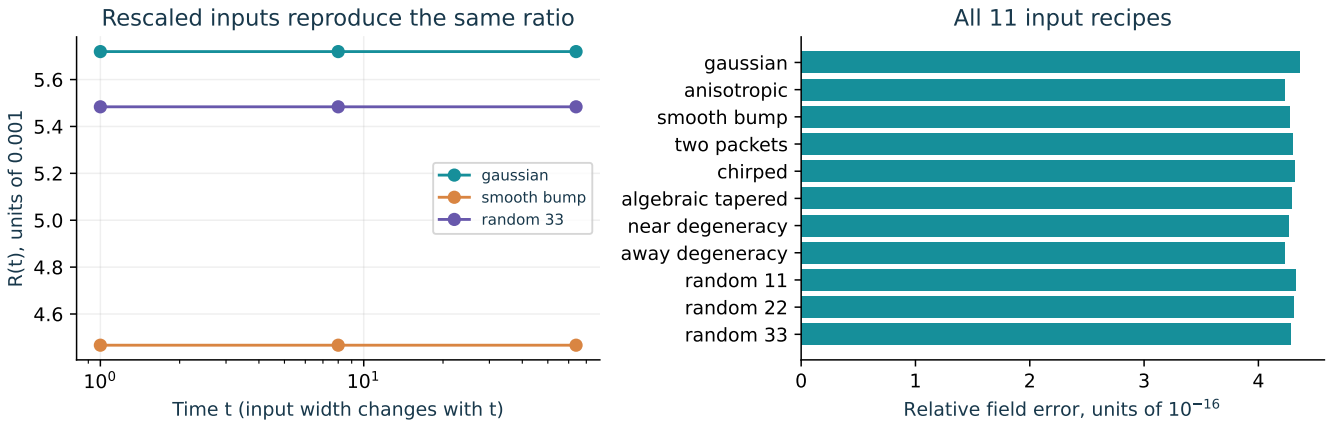
RESEARCH FOLLOW-UP / SEPTEMBER 10, 2026

1. Does the solver preserve cubic scaling?

The literature review suggested replacing a purely visual search with experiments that target the structure of the dispersion. We begin with an exact symmetry of the homogeneous equation, then vary spectral geometry and cancellation, and test an independent integral calculation.

$$f_\lambda(X) = \lambda^3 f(\lambda X), \qquad \|f_\lambda\|_1 = \|f\|_1, \qquad R(t; f_\lambda) = R(\lambda^3 t; f).$$

Here $X = (x, y, z)$ and $\beta = 0$. At $t = 8/\lambda^3$, all three rescaled inputs should produce the same compensated ratio as the original at $t = 8$.



Read this first. All eleven recipes pass the scaling check. Across 33 evaluations, the largest relative complex-field discrepancy is `4.4e-16`.

The calculation uses $\lambda = 1/2, 1, 2$, hence times $64, 8, 1$. A length-144, 96^3 grid is rescaled to length $144/\lambda$ and spacing $1.5/\lambda$. Initial data and the dispersion symbol are reevaluated on each scaled grid. The left panel shows three representative recipes; the right summarizes all eleven.

What this verifies. This is a check of the discrete implementation of an exact symmetry. The matched grids make the relation exact even for a finite periodic model; machine-precision agreement is not a new whole-space error bound.

What scaling says about the research question. If a bound with exponent γ and the same L^1 right-hand side held uniformly for every input at all $t \geq 1$, then at $t = \lambda^{-3}$ it would require

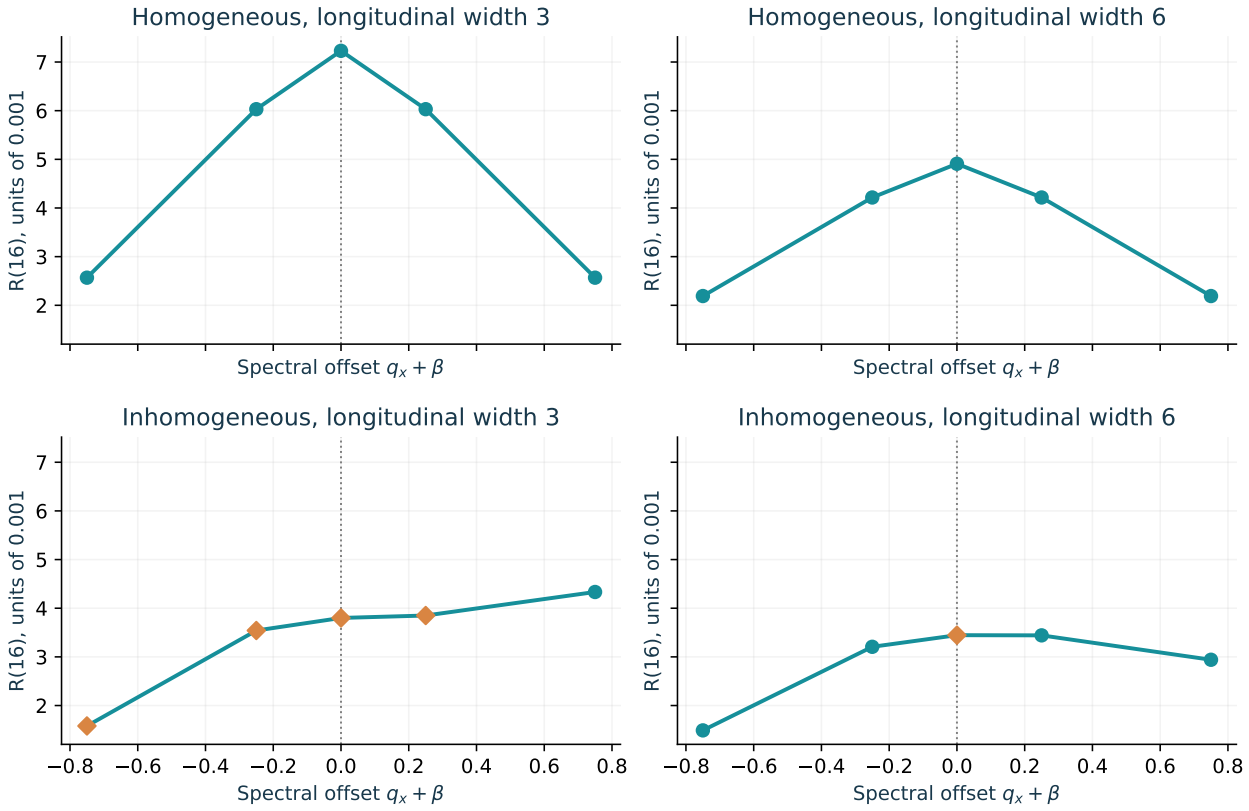
$$\lambda^{7/2} \big\| |D_x|^{1/2} S(1) f \big\|_\infty \leq C \lambda^{3\gamma} \|f\|_1.$$

For a nonzero weighted solution this is impossible as $\lambda \to 0$ when $\gamma > 7/6$. Improving the homogeneous result therefore requires a different question: a better constant, or faster decay under additional input assumptions. This derivation does not apply unchanged to fixed $\beta = 1$.

RESEARCH FOLLOW-UP / TARGETED FREQUENCIES

2. Where the spectrum sits changes the ratio

We vary longitudinal width $\sigma_x \in \{3, 6\}$ and spectral offset $q_x + \beta \in \{-0.75, -0.25, 0, 0.25, 0.75\}$, holding transverse width at 3. These give twenty cases. Three more target the second curvature-degenerate set, two impose zero spatial integral, and one supplies a Gaussian cancellation control: 26 inputs, each evaluated at $t = 4, 8, 16$.



Reading the markers. Circles use the fine 3D FFT result after passing the 2% screen. Diamonds use a separate Gaussian-integral follow-up after the initial screen failed. The dotted line marks the multiplier's cusp.

Read this first. The homogeneous ratios are largest at the central frequency among these sampled offsets. The inhomogeneous patterns differ. Frequency location and width both matter; there is no single ordering that transfers across regimes.

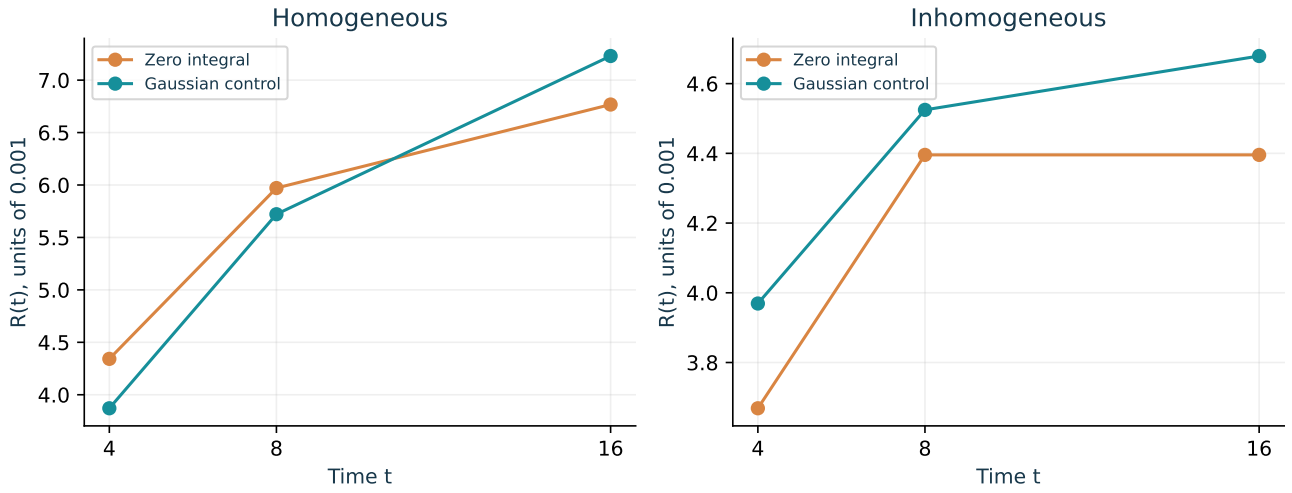
The original checks compare lengths 144 and 192 at spacing 1.5, then spacings 1.5 and 1 at length 192. 70 of 78 case–time measurements pass the 2% screen. Eight are flagged and recomputed using analytic transverse integration, periods 8192 and 16384, radial limits 256 and 512, and local integral optimization. For the odd Gaussian control, the Fourier amplitude is multiplied by $-i\sqrt{\pi/2}\,\sigma_x k$ to enforce exact continuum $L^1 = 1$ normalization; the other follow-ups use normalized Gaussians. 8 of eight pass the 0.1% follow-up screen. The largest change between those two broad searches is 0.0011%; the largest correction to an original fine-grid peak is 2.288%.

Second degeneracy. For the inhomogeneous symbol we also test $q_x = 0.5$, $q_y = \sqrt{0.75} + \delta$ with $\delta = -0.2, 0, 0.2$ and widths 6. The middle center lies on $q_x(q_x + 1) = q_y^2$. These are localized Gaussian spectra, not exact support on the surface. All three pass the original screen at $t = 16$; the central case is not the largest of the three. This small test does not identify a worst-case input.

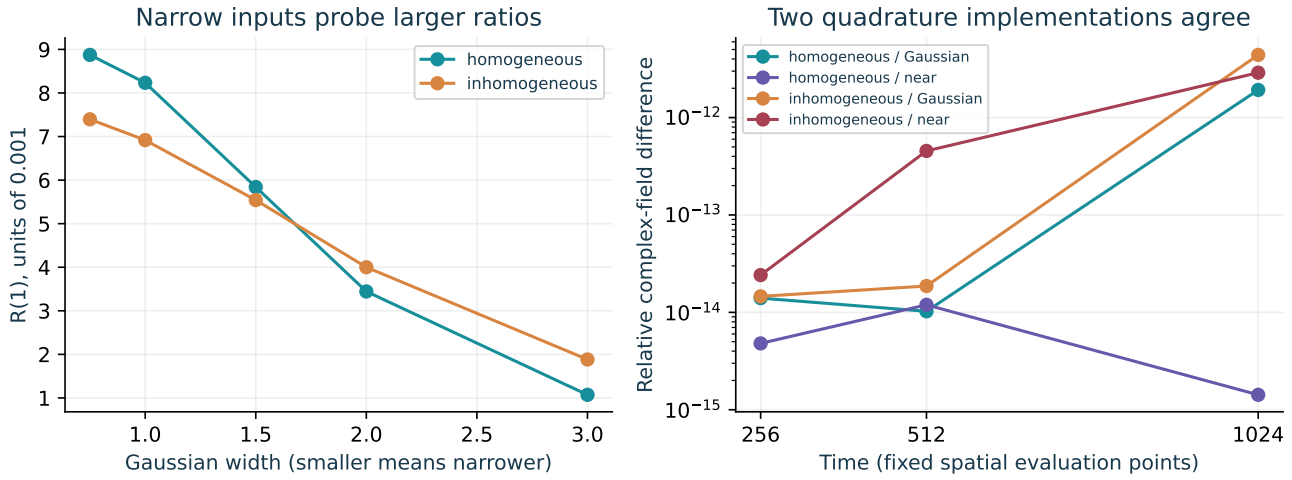
RESEARCH FOLLOW-UP / CANCELLATION AND CONCENTRATION

3. Test assumptions, not just different shapes

For each regime, compare an unshifted Gaussian with $f(X) \propto x \exp[-|X|^2/18]$, normalized separately to $L^1 = 1$. The latter has zero spatial integral by oddness: low-frequency cancellation is built into the input rather than inferred from a picture.



Interpretation. These are short-time ratios, not fitted asymptotic rates. The inhomogeneous zero-integral ratio is almost flat from 8 to 16, within the numerical differences tested here. A zero integral changes the curve but does not, by itself, establish a stronger uniform estimate. These inputs still have Gaussian spectral tails; no experiment with strictly supported spectra away from degeneracies has been completed.



Left: concentrated inputs. At $t = 1$, exactly L^1-normalized Gaussian widths decrease from 3 to 0.75. Ratios increase in both regimes. All ten cases pass a 2% numerical screen: longitudinal period/radial extent are doubled, radial spacing is halved, and candidate peaks are checked by non-periodic integration. The longitudinal spacing is 0.25 and the narrowest width spans three such steps.

Read this first. Narrower inputs probe larger ratios here, but five widths do not establish convergence to a kernel maximum or determine the optimal constant.

Right: independent integration. The comparison is explained on the next page. Its later-time points evaluate fixed coordinates, not newly located maxima.

RESEARCH FOLLOW-UP / INDEPENDENT INTEGRATION

4. A second quadrature agrees through $t = 1024$

We compare two separately implemented methods for the reduced Gaussian integral. Both handle the multiplier cusp at $k = -\beta$, but use different panel placement and integration rules.

Method	How it evaluates the integral
Fixed Gauss–Legendre	64 Gaussian-frequency panels plus a cusp split; $k = -\beta \pm u^2$ regularization; 256 and 512 nodes per panel.
Adaptive Gauss–Kronrod	Panels limit the principal cubic phase variation; stationary points of $tk^3 + xk$ are inserted; the residual complex amplitude is integrated adaptively after a separately implemented cusp substitution.

Read this first. Across twelve pointwise comparisons, the maximum relative complex-field difference is `4.4e-12`. No adaptive integration warnings were recorded. Expanding frequency truncation from 12 to 14 Gaussian standard deviations changes values by at most `7.5e-12` relatively.

For each of the four earlier Gaussian-family cases, we keep its refined $t = 256$ peak coordinates and evaluate the field there at $256, 512, 1024$. The 512-node Gauss–Legendre values agree with the adaptive calculation. For two cases at 1024, changing from 256 to 512 nodes changes the field by about 10^{-7} relatively, showing why integration order needs checking as time increases.

What has advanced. We have verified a scaling identity, completed a controlled frequency sweep with follow-ups for every initially flagged point, compared cancellation inputs, tested a family of concentrated inputs, and checked a second integration implementation at later times.

What has not advanced yet. Global peak searches at 512 and 1024 have not been performed. Their locations can move, so the pointwise comparisons cannot extend the decay curves. The new screens remain numerical consistency checks, not certified bounds on a continuum supremum. No measured fluid data or full nonlinear water-wave simulation is added in this batch.

Research connection. Frequency targeting is motivated by Ben-Artzi, Koch and Saut, *Dispersion Estimates for Third Order Equations in Two Dimensions* (2003). Stationary-point-aware integration is motivated by Huybrechs, Kuijlaars and Lejon, *A Numerical Method for Oscillatory Integrals with Coalescing Saddle Points* (2019), and Gibbs, Hewett and Huybrechs, *Numerical evaluation of oscillatory integrals via automated steepest descent contour deformation* (2024). Our adaptive method is a simpler comparison calculation; it does not implement their published algorithms or claim their error guarantees.

Reproduce. Run `research_experiments.py` with arguments `scaling`, `sweep`, `quadrature`, and `identity`; then run `refine_research_flags.py` and `report/make_research_report.py`. Case-level checkpoints, configurations and summaries are in `results/research_experiments/`.

The annotated reading note is `research/literature_review.md`.